
Adaptive Peak Shaving via Greedy Cap-Ratio: A Feasibility-Guarded ESS Scheduling Strategy for Residential Demand-Charge Reduction

Abstract

Behind-the-meter energy storage systems (ESS) offer significant potential for electricity bill reduction through energy shifting and, crucially, peak shaving to mitigate demand charges. However, optimal scheduling often relies on computationally expensive solvers and perfect future information, hindering real-time deployment. This paper proposes Greedy Cap-Ratio (GCR), a practical online controller featuring: (i) Mandatory Discharge to enforce a dynamic peak cap, (ii) a Look-ahead Deficit Guard to maintain SOC feasibility by preemptively identifying energy shortages, and (iii) Step-wise No-export Clipping to ensure strict adherence to grid interconnection constraints. Benchmarked against an LP-based “Clairvoyant” Oracle and rule-based heuristics across 11 households (2013-01 and 2013-02), GCR achieved up to 14.0% cost reduction and a consistent 25.0% peak reduction, demonstrating robust generalization to unseen demand patterns at negligible computational overhead.

1. Introduction

Residential electricity tariffs increasingly include demand charges that depend on the maximum grid import within a billing period. Consequently, peak shaving can dominate the value of behind-the-meter ESS compared with pure time-of-use (TOU) energy arbitrage. Optimization-based schedulers can approach global optima under linear dynamics but typically require forecasts (or perfect future information) and repeated solver calls, complicating scalable real-time deployment.

We propose Greedy Cap-Ratio (GCR), a feasibility-guarded online scheduling strategy designed to: (i) explicitly target demand-charge reduction via cap enforcement, (ii) preserve operational feasibility under SOC and power limits, and (iii) guarantee strict compliance with anti-backfeed (no-export) interconnection constraints. We benchmark against a rule-based heuristic and an LP-based clairvoyant oracle that assumes full access to future net-load trajectories.

Contributions: (a) a standardized formulation highlighting demand-charge structure and no-

export feasibility, (b) a novel Look-ahead Deficit Guard for safety-aware energy budgeting, (c) cross-month robustness where a parameter tuned on January data maintains performance on February data (robustness to seasonal variance), and (d) an explicit AI–human co-scientist workflow aligned with the challenge objectives.

2. Related work

Demand-charge-aware ESS control has been studied using convex optimization, model predictive control, and heuristic policies. Optimization methods can provide strong performance but may be brittle under forecast errors and can be expensive at scale. Rule-based controllers are simple but may violate practical constraints or deplete energy prematurely without safeguards. GCR bridges these regimes by retaining greedy simplicity while embedding explicit feasibility guards and strict no-export safety.

3. Data and tariff setting

We evaluate on 11 residential households over two months (2013-01 and 2013-02) with 30-minute resolution ($\Delta t = 0.5$ h). The tariff

combines TOU energy prices and a demand charge proportional to monthly peak grid

Table 1: Simulation parameters and constraints.

Item	Value
Time step Δt	0.5 h
Battery capacity	10 kWh
SOC bounds	[0.1, 0.9]
Initial SOC	0.5
Max charge/discharge	3 kW / 3 kW
Efficiencies	$\eta_c = 0.95, \eta_d = 0.95$
No-export	$p_{\text{grid}} \geq 0$
Demand charge	12,000 KRW/kW (configurable)

4. Problem formulation

Let $P_{\text{load}}(t)$ denote household load and $P_{\text{pv}}(t)$ denote local PV generation at time $t \in \{1, \dots, T\}$. Net load is $P_{\text{net}}(t) = P_{\text{load}}(t) - P_{\text{pv}}(t)$. Decision variables are charging power $P_{\text{ch}}(t) \geq 0$ and discharging power $P_{\text{dis}}(t) \geq 0$. Grid import follows:

$$P_{\text{grid}}(t) = P_{\text{net}}(t) + P_{\text{ch}}(t) - P_{\text{dis}}(t).$$

Constraints include:

No-export: $P_{\text{grid}}(t) \geq 0, \forall t$.

SOC dynamics: $\text{SOC}(t+1) = \text{SOC}(t) + \left(\eta_c P_{\text{ch}}(t) - \frac{1}{\eta_d} P_{\text{dis}}(t) \right) \Delta t$.

Bounds: $\text{SOC}_{\min} \leq \text{SOC}(t) \leq \text{SOC}_{\max}$ and $0 \leq P_{\text{ch}}(t) \leq P_{\text{ch}}^{\max}$, $0 \leq P_{\text{dis}}(t) \leq P_{\text{dis}}^{\max}$.

Monthly objective (energy + demand charge):

$$J = \sum_{t=1}^T \left(\text{Price}(t) P_{\text{grid}}(t) \Delta t + \lambda \max_{t \in \{1, \dots, T\}} P_{\text{grid}}(t) \right)$$

4.1 Safety properties

Proposition 1 (No-export safety). The step-wise clipping rule in GCR guarantees $P_{\text{grid}}(t) \geq 0$ for all t .

Proof sketch. After computing tentative $(P_{\text{ch}}, P_{\text{dis}})$, GCR enforces $P_{\text{dis}}(t) \leftarrow \min\{P_{\text{dis}}(t), P_{\text{net}}(t) + P_{\text{ch}}(t)\}$. Substitution yields $P_{\text{grid}}(t) = P_{\text{net}}(t) + P_{\text{ch}}(t) - P_{\text{dis}}(t) \geq 0$ by construction.

Proposition 2 (Cap enforcement under feasibility). If sufficient discharge power and energy are available (SOC and $P_{\text{dis}}^{\max}$)

import. ESS parameters and tariff coefficients are summarized in Table 1.

not binding), Mandatory Discharge enforces $P_{\text{grid}}(t) \leq \text{Cap}$ whenever $P_{\text{net}}(t) > \text{Cap}$.

5. Greedy Cap-Ratio (GCR)

GCR defines a peak cap $\text{Cap} = r \cdot P_{\text{peak}}^{\text{base}}$, where $P_{\text{peak}}^{\text{base}} = \max_t P_{\text{net}}(t)$ and $r \in (0, 1]$ is a cap ratio. With $r = 0.75$, the target cap corresponds to a 25% reduction relative to baseline peak. GCR executes Mandatory Discharge, Look-ahead Deficit Guard, and Step-wise No-export Clipping at each time step.

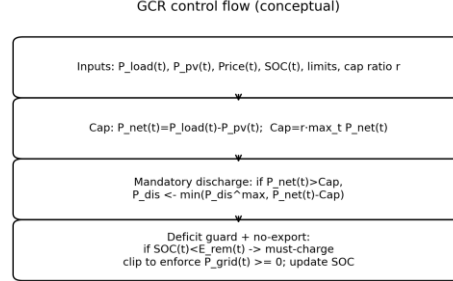


Figure 1: GCR control flow (conceptual).

5.1 Mandatory discharge

When $P_{\text{net}}(t) > \text{Cap}$, mandatory discharge is:

$$P_{\text{dis}}^{\text{mand}}(t) = \min\left(P_{\text{dis}}^{\max}, P_{\text{net}}(t) - \text{Cap}\right).$$

5.2 Look-ahead deficit guard (energy budgeting)

GCR incorporates safety-aware energy budgeting to avoid premature depletion. Define the mandatory discharge energy at time k as $E_{\text{mand}}(k) = \max(0, P_{\text{net}}(k) - \text{Cap}) \cdot \frac{\Delta t}{\eta_d}$. The remaining mandatory energy is the suffix sum:

$$E_{\text{rem}}(t) = \sum_{k=t}^T \max(0, P_{\text{net}}(k) - \text{Cap}) \cdot \frac{\Delta t}{\eta_d}.$$

If $\text{SOC}(t) < E_{\text{rem}}(t)$, the controller enters a Deficit State and must-charge irrespective of the current TOU price, ensuring the cap can be defended in later intervals.

5.3 Step-wise no-export clipping

To ensure strict anti-backfeed compliance, GCR clips discharge whenever

$P_{\text{net}}(t) + P_{\text{ch}}(t) - P_{\text{dis}}(t) < 0$ by setting $P_{\text{dis}}(t) \leftarrow P_{\text{net}}(t) + P_{\text{ch}}(t)$.

5.4 Computational complexity

GCR runs in $\mathcal{O}(T)$ time per household-month. With precomputed suffix sums, each step uses constant-time arithmetic and clipping. The LP oracle, by contrast, requires solving a month-scale convex optimization per household.

6. Experimental protocol

Baselines: BASE (no ESS), RULE (heuristic TOU charging + peak-time discharge), and LP (clairvoyant oracle with perfect future knowledge). Cap-ratio tuning: we sweep r in $\{0.95, 0.90, 0.87, 0.85, 0.83, 0.80, 0.78, 0.75\}$ on January and select $r=0.75$ as a robust operating point. We then evaluate on February without retuning to test robustness to seasonal variance.

Table 2: Mean monthly total cost and peak demand across 11 households.

Month	BASE total	RULE total	OURS total	LP total	BASE peak	RULE peak	OURS peak	LP peak
2013-01	92,974.4	86,681.6	80,319.7	61,861.9	3.368	2.884	2.526	1.349
2013-02	88,898.4	80,278.8	76,447.8	58,065.4	3.420	2.739	2.565	1.296

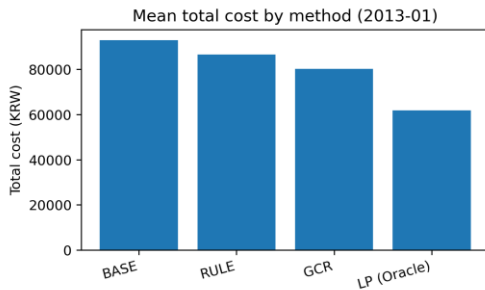


Figure 3: Mean total cost by method (2013-01).

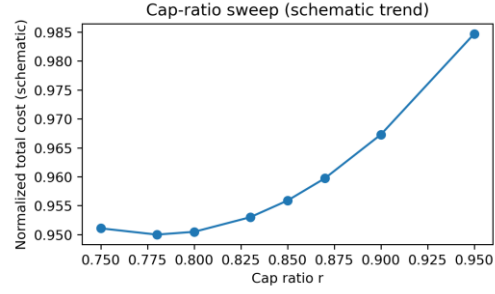


Figure 2: Cap-ratio sweep (schematic; qualitative tuning trend).

7. Results

GCR reduces mean total cost by 13.6% (Jan) and 14.0% (Feb) relative to BASE, and by 7.3% (Jan) and 4.8% (Feb) relative to RULE (Table 2). Peak reduction is exactly 25.0% in both months, consistent with the enforced cap ratio $r=0.75$.

Peak reduction as an enforced upper bound. Because $\text{Cap}=0.75 \cdot P_{\text{peak}}^{\text{base}}$, Mandatory Discharge constrains peak import by design (subject to physical limits). Robustness to seasonal variance. The same tuned r maintains identical peak reduction and similar cost savings on February without retuning.

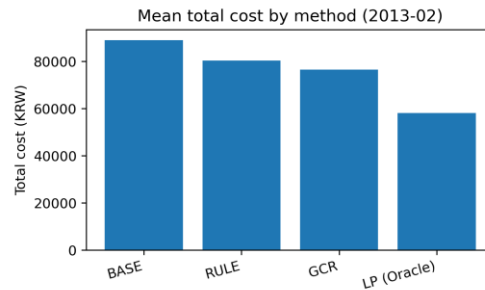


Figure 4: Mean total cost by method (2013-02).

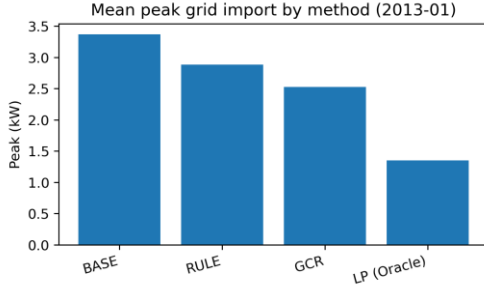


Figure 5: Mean peak grid import by method (2013-01).

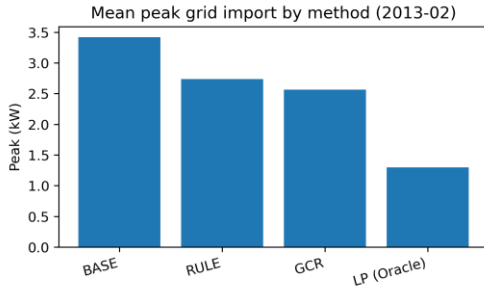


Figure 6: Mean peak grid import by method (2013-02).

Table 3: Household-level wins of GCR over RULE.

Month	Count(OURS total < RULE)	Count(OURS peak < RULE)
2013-01	8/11	7/11
2013-02	9/11	7/11

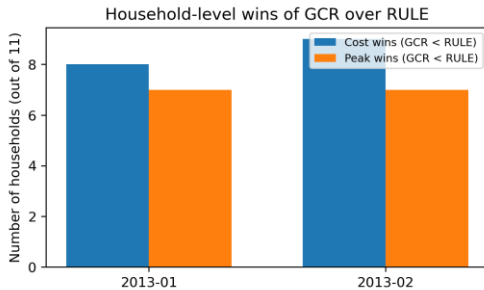


Figure 7: Household-level wins of GCR over RULE (out of 11).

8. Discussion

Why the Deficit Guard matters. Peak-triggered discharge alone may deplete the battery early, failing to defend late-month peaks. By budgeting the remaining mandatory discharge energy via $E_{\text{rem}}(t)$, GCR anticipates shortfalls and recovers SOC proactively.

A. Appendix: Greedy Cap-Ratio pseudocode

Inputs: $P_{\text{load}}(t)$, $P_{\text{pv}}(t)$, $\text{Price}(t)$, limits ($\text{SOC}_{\text{min}}/\text{max}$, $P_{\text{ch}}^{\text{max}}$, $P_{\text{dis}}^{\text{max}}$), efficiencies η_{c} , η_{d} , cap ratio r

Oracle gap interpretation. The LP oracle is clairvoyant and can allocate energy globally to future peaks, serving as an upper bound on performance. GCR trades optimality for provable safety, negligible overhead, and deployability.

No-export compliance. The clipping step directly enforces anti-backfeed constraints common in residential interconnections without additional optimization.

9. Limitations

First, suffix-sum budgeting requires future net-load estimates; deployment can approximate $E_{\text{rem}}(t)$ with short-horizon forecasts or conservative profiles. Second, the cap may be violated if hardware limits bind; GCR mitigates risk via the Deficit Guard but cannot exceed physical constraints. Third, we report monthly means; future work will add dispersion measures and confidence intervals.

10. Broader impacts

Positive impacts include lower bills and reduced peak demand, supporting grid reliability. Potential negative impacts include inequitable access due to upfront ESS costs; policy mechanisms may be needed to broaden access. The method uses meter/PV time series only and introduces no direct privacy risks.

11. AI-human co-scientist synergy

The development of GCR resulted from iterative collaboration with an AI Co-Scientist. Heuristic derivation: the AI proposed the Deficit Guard after diagnosing failures where ESS energy was depleted before late-month peaks. Formal verification: the AI helped translate the control logic into a robust Python simulation and verified no-export clipping consistency. Generalization analysis: AI-driven sensitivity analysis over r identified a robust operating point ($r=0.75$) balancing peak shaving and SOC sustainability across heterogeneous households.

References

- [1] S. Boyd and L. Vandenberghe. Convex Optimization. Cambridge University Press, 2004.
- [2] S. Diamond and S. Boyd. CVXPY: A Python-Embedded Modeling Language for Convex Optimization. JMLR, 2016.

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Compute: P_net(t)=P_load(t)-P_pv(t); P_peak^base=max_t P_net(t);
Cap=r·P_peak^base
E_mand(t)=max(0,P_net(t)-Cap)·Δt/η_d; E_rem(t)=suffix-
sum(E_mand)
For t=1..T:
  if P_net(t)>Cap: P_dis <- min(P_dis^max, P_net(t)-Cap) else P_dis <- 0
  if SOC(t)<E_rem(t): P_ch <- min(P_ch^max, (SOC_max-SOC(t))/(η_c·Δt)) #
must-charge
  else: P_ch <- optional price-gated charging
  if P_net(t)+P_ch-P_dis < 0: P_dis <- P_net(t)+P_ch # no-export
clipping
  SOC(t+1) <- clip(SOC(t)+η_c P_ch Δt - (1/η_d)P_dis Δt, SOC_min,
SOC_max)

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AI Co-Scientist Challenge Korea paper checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The abstract and Section 1 state the three components and report reductions supported by Table 2 and Figures 3–6.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 9 covers forecast dependence, hardware-limit binding cases, and missing variance statistics.

3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper provides safety propositions with proof sketches (Section 4.1) but no formal theorems requiring full proofs.

4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: Sections 4–7, Table 1, and Appendix A specify the algorithm, constraints, and protocol.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

Answer: [No]

Justification: We do not claim public release of the competition-provided dataset or code at submission time.

6. Experimental Setting/Details

Question: Does the paper specify all experimental details (splits, hyperparameters, how chosen) needed to understand results?

Answer: [Yes]

Justification: Section 6 describes tuning on January via cap-ratio sweep and evaluation on February without retuning.

7. Experiment Statistical Significance

Question: Does the paper report error bars or other significance information?

Answer: [No]

Justification: Results are month-level means across 11 households; future work will add dispersion measures and confidence intervals.

8. Experiments Compute Resources

Question: Does the paper provide sufficient information on compute resources needed to reproduce the experiments?

Answer: [Yes]

Justification: GCR is $O(T)$ and runs CPU-only; LP oracle requires a standard convex solver (Section 5.4).

9. Code Of Ethics

Question: Does the research conform with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: The work uses non-invasive meter data and does not involve human subjects; anonymity is preserved.

10. Broader Impacts

Question: Does the paper discuss positive and negative societal impacts?

Answer: [Yes]

Justification: Section 10 covers benefits (peak reduction) and risks (inequitable access).

11. Safeguards

Question: Does the paper describe safeguards for responsible release of high-risk data/models?

Answer: [N/A]

Justification: The submission does not release high-risk models or scraped datasets.

12. Licenses for existing assets

Question: Are creators/owners properly credited and are licenses/terms of use mentioned and respected?

Answer: [N/A]

Justification: We do not redistribute the dataset; core software dependencies are cited in References.

13. New Assets

Question: Are new assets introduced well documented and provided alongside the assets?

Answer: [N/A]

Justification: No new public assets are introduced; the algorithm is documented in Appendix A.

14. Crowdsourcing and Research with Human Subjects

Question: For crowdsourcing/human subjects, are instructions and compensation details included?

Answer: [N/A]

Justification: No crowdsourcing or human-subject experiments were conducted.

15. Institutional Review Board (IRB) Approvals

Question: Are risks and IRB approvals described for human subjects research?

Answer: [N/A]

Justification: Not applicable because the work does not involve human-subject interventions.